

IDP Login Performance Testing: Tool Evaluation, Workarounds, and Strategic Recommendations

Overview

This document provides an in-depth evaluation of three performance testing strategies for **IDP Login flows**, including:

- JMeter
- Internal Selenium Grid
- BlazeMeter with Taurus

The goal is to determine the most viable, scalable, and technically suitable approach to simulate IDP login traffic — considering JavaScript-heavy redirects, token/cookie handling, CS pages, and security limitations.

Use Case Background

IDP login flows include multiple dynamic redirects, session token generations, JavaScript-based authentication functions (e.g., submit, dispatch, servlet), and browser-managed behaviors. These flows challenge traditional load testing tools that simulate only protocol-level (HTTP) behavior.

We are currently:

- Using **JMeter** for API-level load tests (partial success).
- Exploring **browser-based simulation** to fully replicate authentication logic.

Challenge Summary

Issue	Impact Level	Notes
JavaScript Execution Required	High	JMeter cannot execute JS; breaks flow at CS page/servlet level
Session/Token Handling	Medium	Requires capturing dynamic tokens/cookies manually in JMeter
Redirects + DOM Events	High	CS pages behave like browser apps — hard to mimic in raw HTTP
CAPTCHA / MFA	High	Requires browser-based logic and/or workarounds
Firewall / Network Barriers	Medium	BlazeMeter requires whitelisting IPs or setting up private location

Tool Options Evaluation

1. JMeter

Protocol-based simulation. Works for basic login flows without JS.

Pros:

- Easy to scale.
- Cost-effective (open source).
- Existing suite available.
- Integrates well with CI/CD (Jenkins, Azure DevOps).

Cons:

- No JS execution.
- Manual token/cookie extraction.
- Cannot handle redirect logic based on DOM events.
- CAPTCHA/MFA unsupported.

Workarounds:

- Record login via browser → capture tokens → inject via CSV.
- Chain Selenium script (for session init) → pass session to JMeter.

Best For: Load testing backend APIs or simplified IDP flows.

2. Selenium Grid (Internal)

Browser-based simulation using Chrome/Firefox.

Pros:

- Handles all JavaScript, token flows, DOM events.
- Can mimic end-to-end user behavior.
- Can solve redirect/session logic naturally.

Cons:

- Low scalability: max 30–60 users (based on nodes).
- Not designed for performance testing.
- Resource-heavy (RAM/CPU per browser).
- Reporting limitations (latency/TPS/SLA).

Workarounds:

- Use for *functional validation* and token generation.
- Extract token/cookie/session pass to JMeter.
- Combine with Taurus for semi-automated runs.

Best For: Functional simulation of Web Reg/IDP/MFA for up to ~60 users.

3. BlazeMeter (Taurus/Selenium/Cloud) *(Time Consuming so not Applicable for the timing)*

Cloud-based load testing with real browser or API simulation.

Pros:

- Browser load at scale (1000s of users).
- Supports JS, tokens, cookies, redirects.
- Simulates real user behavior.
- CI/CD friendly (via Taurus YAML).

Cons:

- Pages behind firewall needs IP whitelisting or private location setup.
- Dedicated IPs cost extra.
- Limited by license (VUH: Variable Unit Hours).
- Debugging more complex.

Workarounds:

- Move CS page behind external proxy or expose a staging clone.
- Use Private Location setup (if license allows).
- Run initial functional test via Grid, load via BlazeMeter.

Best For: Scalable browser-based load of external endpoints.

Hybrid Strategy (Recommended)

Layer	Tool	Status	Purpose	Note
IDP Login using JS & CS pages	JMeter + Blazemeter	Ready to use	Login + Capture session/cookies	This is been used for 100 users in Stage 1
Selenium + Taurus Browser simulation	Selenium + Taurus	In progress	Hit backend with 100s–1000s users	Completed for 1 account need to add more & test
Cloud UI Load	BlazeMeter	Not Required	UI load (if firewall allows), Research Item & has learning curve	Research item & time consuming due whitelisting IPs etc. so skipping this at this point

Strategic Recommendations

Scenario	Tool	Notes
Simple IDP Login via API & some CS pages	JMeter	Already built, use for scale
Web Registration / Token DOM Events	Selenium	Needs new test suite
High-scale UI Load	BlazeMeter	Coordinate IP setup, Research item & has learning curve, N/A
Real-user experience validation (30–60)	Selenium + Taurus	Internal grid use. in progress

Action Items

1. Continue using JMeter for basic IDP login testing.
2. Build Selenium test for full UI flow (IDP Login & Web Registration).
3. Coordinate with infra/security team on BlazeMeter: *(time consuming)*
 - IP whitelisting
 - Dedicated IP pricing
 - Private location rights
4. Report hybrid plan to QA leadership & product owners.